

4.7 Exploring the AWS Well-Architected Framework

In this course, we will:

- Outline the value proposition of the AWS Cloud and understand the economies of scale
- Describe the AWS global infrastructure
- List the advantages of high availability, elasticity, and agility
- Define the AWS Well-Architected Framework and the six pillars of design principles

The Value Proposition of the AWS Cloud

What Is Cloud Computing?

- "Cloud computing is the on-demand delivery of compute power, database, storage, applications, and other IT resources through a cloud services platform via the internet with pay-as-you-go pricing
- Whether you are running applications that share photos to millions of mobile users or you're supporting the critical operations of your business, a cloud services platform provides rapid access to flexible and low-cost IT resources"

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/what-is-cloud-computing.html>



What Is Cloud Computing?

- "With cloud computing, you don't need to make large upfront investments in hardware and spend a lot of time on the heavy lifting of managing that hardware
- Instead, you can provision exactly the right type and size of computing resources you need to power your newest bright idea or operate your IT department
- You can access as many resources as you need, almost instantly, and only pay for what you use"

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/what-is-cloud-computing.html>



Trade Fixed Expense for Variable Expense

- **Only pay for what is needed and used**
- Customers can pay only when they consume computing resources
- The default model is to only pay for what is consumed, much like a utility bill
- Reduce upfront capital expenditures by avoiding heavy investments in server farms, data centers, and blade server stacks before you know how you are going to use them





Benefit from Massive Economies of Scale

- By using cloud computing, customers can obtain a much lower adaptable cost than they can get on their own
- New customers can leverage the utilization and experience of hundreds of thousands of other customers aggregated in the cloud
- Cloud service providers like AWS can realize higher economies of scale, which leads to lower "pay as-you-go" pricing

Stop Guessing Capacity

- Remove the need to guess about infrastructure and service capacity needs
- Customers typically find themselves either sitting on costly idle resources or struggling with inadequate capacity
- Proper capacity decisions can be made prior to application deployment
- A cloud consumer can access as much or as little capacity as needed, and scale up and down as required, with only a few minutes notice





Increase Speed and Agility

- Cutting-edge IT resources are only a click away in a cloud computing environment
- Customers decrease the time to make resources available to administrators and developers from weeks to merely minutes
- The outcome of cloud computing is a substantial increase in agility for the enterprise
 - The resources (cost and time) necessary to experiment and develop are considerably lower

Stop Spending Money Running and Maintaining Data Centers

- Cloud customers can prioritize projects and propositions that set the business apart from competitors rather than the infrastructure
- AWS cloud computing lets organizations focus on their own customers instead of the overhead and heavy lifting of racking, stacking, and powering racks of server blades
- Large and unsustainable data centers will become a thing of the past due to global cloud consuming



Go Global in Minutes



- Organizations can effortlessly deploy data and applications in several regions around the world with just a few simple clicks
 - This results in lower latency and a better experience for customers at a minimal cost
- AWS customers are leveraging edge locations and availability zone data centers in over 30 geographic regions throughout the world
- New availability zones and regions are added every year

Economies of Scale

Economies of Scale

- This concept refers to the ability to lower costs and raise efficiency when operating at a larger scale in comparison to operating at a smaller scale
- Since cloud customers are becoming more motivated and willing to drive growth through technology, they often look to strategies such as cost-reducing while enabling innovation

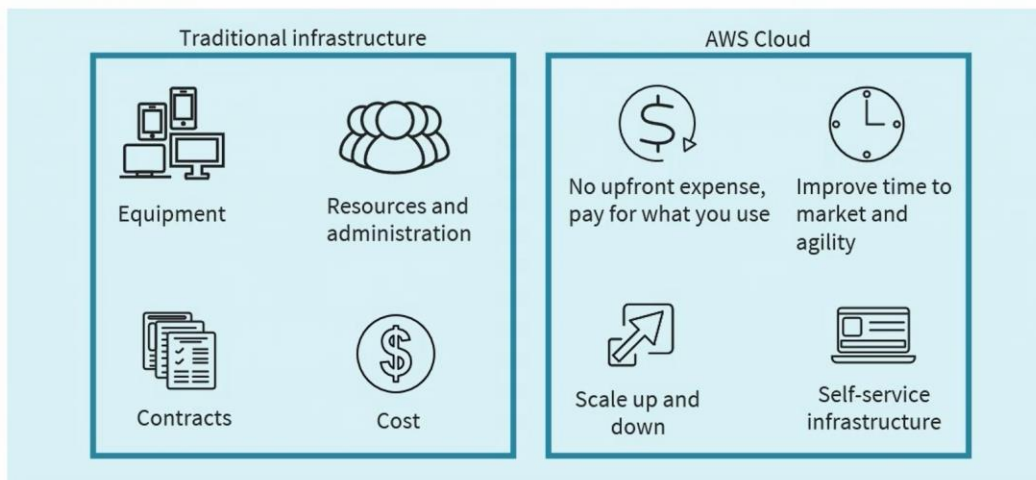


Economies of Scale



- Cloud economics transcends simply cutting cloud costs – it is about addressing business goals through greater speed, agility, and flexibility
- Considering the larger perspective in this way helps IT groups to select the best cloud solution for their projects, programs, and initiatives

Economies of Scale



Economies of Scale

Types of economies of scale	
Technical	Large capital equipment with high fixed costs
Specialization	Division of labor and specialization within production – more efficient with high output
Bulk buying	Lower average costs for buying large quantity
Marketing	National ad campaign more efficient for high sales
Risk bearing	Bigger firms more able to survive downturns
Container principle	Increase in surface area leads to double the increase in volume
Financial	Large firm gets better rate of interest from bank
External	When a firm benefits from the whole industry Increasing in size

"Definition of economies of scale" Via Economics Help.org, <https://www.economicshelp.org/microessays/costs/economies-scale>

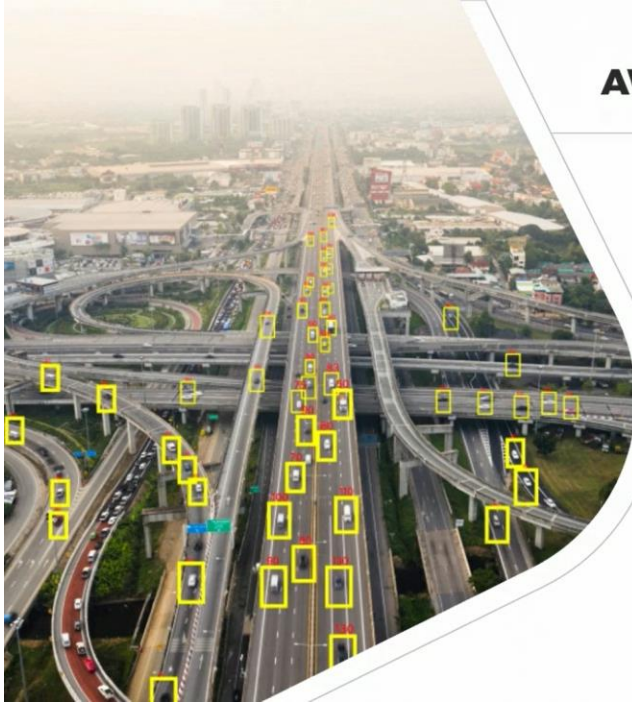
Exploring the AWS Global Infrastructure

Advantages of High Availability, Elasticity and Agility

Advantages of High Availability

- Protect against data center, Availability Zone (AZ), server, network, and storage subsystem failures
 - Keep your organization or business running with little or no downtime
- All AZs in an AWS region are interconnected with high-bandwidth, low-latency networking, over fully redundant, dedicated metro fiber
- Data centers located in different AWS AZs have a discrete, uninterruptible power supply and onsite backup generation facilities





AWS High Availability (HA)

- Most providers of real-time communications align with service levels that provide availability from 99.9% to 99.999%
- Depending on the degree of HA needed at AWS, customers can take progressively sophisticated steps along the full life cycle of the service or application

AWS HA

- AWS recommends following these guidelines to achieve a robust degree of high availability:
 - Design the system or application to have no single point of failure by using automated monitoring, detection, and failover mechanisms for both stateless and stateful components
 - Correctly instrument and test system availability
 - Prepare operating procedures for manual mechanisms to respond to, mitigate, and recover from the failure



Advantages of Elasticity



- Elasticity provides the ability to almost instantly provision and de-provision assorted cloud resources
 - Virtual instances, containers, appliances, database tables, and more
- It involves leveraging dynamic auto-scaling technologies
- Challenges with predicting demand leads to higher costs for the enterprise
- Recent circumstances have revealed how de-provisioning may be the more vital aspect of elasticity

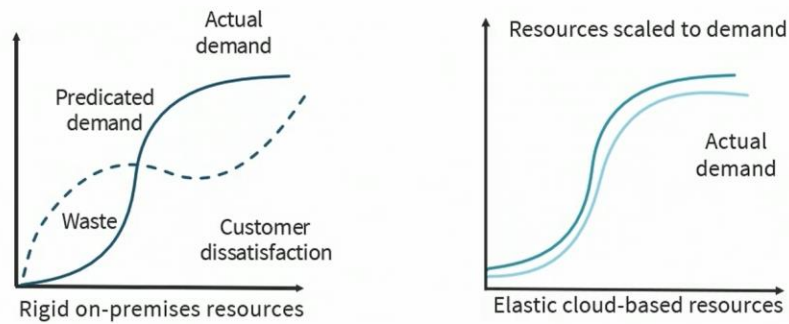
Advantages of Elasticity

"Elasticity of the cloud allows us to add thousands of virtual servers and petabytes of storage within minutes, making such an expansion possible...Leveraging multiple AWS cloud regions, spread all over the world, enables us to dynamically shift around and expand our global infrastructure capacity, creating a better and more enjoyable streaming experience for Netflix members wherever they are"

– Yury Izrailevsky, VP Cloud and Platform Engineering, Netflix (from Netflix Media Center)



Advantages of Elasticity



Advantages of Agility



- Agility leverages features for rapid deployment, testing, experimentation, and innovation
- Customers can quickly overcome geographical limitations
- Content creators can get customer content as close to the consumer as possible
- Agility means reducing time and costs for testing, innovation, and experimentation

Advantages of Agility

- With AWS as the producer, consumers can co-create value and deliver data, applications, services, and solutions
- Agility includes the concept of flexibility in terms of many different options
- The decision to "build or buy" can be more adaptable for DevOps programs
- This value proposition integrates with Agile development and project management



Exploring the AWS Well-Architected Framework

A screenshot of the AWS Well-Architected Framework page. The page has a dark blue header with the AWS logo and navigation links. Below the header, there are six pillars of the framework, each with a title, a brief description, and links to HTML and Labs. The pillars are: Operational Excellence Pillar, Security Pillar, Reliability Pillar, Performance Efficiency Pillar, Cost Optimization Pillar, and Sustainability Pillar.

Operational Excellence Pillar

The operational excellence pillar focuses on running and monitoring systems, and continually improving processes and procedures. Key topics include automating changes, responding to events, and defining standards to manage daily operations.

[HTML](#) | [Labs](#)

Security Pillar

The security pillar focuses on protecting information and systems. Key topics include confidentiality and integrity of data, managing user permissions, and establishing controls to detect security events.

[HTML](#) | [Labs](#)

Reliability Pillar

The reliability pillar focuses on workloads performing their intended functions and how to recover quickly from failure to meet demands. Key topics include distributed system design, recovery planning, and adapting to changing requirements.

[HTML](#) | [Labs](#)

Performance Efficiency Pillar

The performance efficiency pillar focuses on structured and streamlined allocation of IT and computing resources. Key topics include selecting resource types and sizes optimized for workload requirements, monitoring performance, and maintaining efficiency as business needs evolve.

Cost Optimization Pillar

The cost optimization pillar focuses on avoiding unnecessary costs. Key topics include understanding spending over time and controlling fund allocation, selecting resources of the right type and quantity, and scaling to meet business needs without overspending.

Sustainability Pillar

The sustainability pillar focuses on minimizing the environmental impacts of running cloud workloads. Key topics include a shared responsibility model for sustainability, understanding impact, and maximizing utilization to minimize required resources and reduce downstream impacts.

Examining the Well-Architected Framework Design Principles

The screenshot displays the AWS Well-Architected Framework Design Principles page. The browser address bar shows the URL: <https://aws.amazon.com/blogs/apn/the-6-pillars-of-the-aws-well-architected-framework/>. The page features the AWS logo and navigation links: Products, Solutions, Pricing, Documentation, Learn, Partner Network, AWS Marketplace, Customer Enablement, Events, and Explore More. A prominent orange button labeled "Create an AWS Account" is visible in the top right corner. Below the navigation bar, the page title "The 6 Pillars of the AWS Well-Architected Framework" is displayed. The main content area is titled "1. Operational Excellence" and includes a paragraph explaining the pillar's focus on supporting development, running workloads effectively, and improving supporting processes. A link to the "Operational Excellence Pillar whitepaper" is provided. Below this, the "Design Principles" section lists five principles for operational excellence in the cloud: Perform operations as code, Make frequent, small, reversible changes, Refine operations procedures frequently, Anticipate failure, and Learn from all operational failures. The "Best Practices" section states that operations teams need to understand business and customer needs to support business outcomes. A sidebar on the right offers a free 4-week program to prepare for Cloud Practitioner Certification, with a "Register now" link.

1. Operational Excellence

The Operational Excellence pillar includes the ability to support development and run workloads effectively, gain insight into their operation, and continuously improve supporting processes and procedures to delivery business value. You can find prescriptive guidance on implementation in the [Operational Excellence Pillar whitepaper](#).

Design Principles

There are five design principles for operational excellence in the cloud:

- Perform operations as code
- Make frequent, small, reversible changes
- Refine operations procedures frequently
- Anticipate failure
- Learn from all operational failures

Best Practices

Operations teams need to understand their business and customer needs so they can support business outcomes. Ops creates and uses procedures to respond to operational events, and validates their effectiveness to support business

Join our free 4-week program to prepare for Cloud Practitioner Certification
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3. Reliability

The Reliability pillar encompasses the ability of a workload to perform its intended function correctly and consistently when it's expected to. This includes the ability to operate and test the workload through its total lifecycle. You can find prescriptive guidance on implementation in the [Reliability Pillar whitepaper](#).

Design Principles

There are five design principles for reliability in the cloud:

- Automatically recover from failure
- Test recovery procedures
- Scale horizontally to increase aggregate workload availability
- Stop guessing capacity
- Manage change in automation

Best Practices

Before building any system, foundational requirements that influence reliability should be in place. For example, you must have sufficient network bandwidth to your data center. These requirements are sometimes neglected (because they are beyond a single project's scope). With AWS, however, most of the foundational requirements are already incorporated

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4. Performance Efficiency

The Performance Efficiency pillar includes the ability to use computing resources efficiently to meet system requirements, and to maintain that efficiency as demand changes and technologies evolve. You can find prescriptive guidance on implementation in the [Performance Efficiency Pillar whitepaper](#).

Design Principles

There are five design principles for performance efficiency in the cloud:

- Democratize advanced technologies
- Go global in minutes
- Use serverless architectures
- Experiment more often
- Consider mechanical sympathy

Best Practices

Take a data-driven approach to building a high-performance architecture. Gather data on all aspects of the architecture, from the high-level design to the selection and configuration of resource types.

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https://aws.amazon.com/blogs/apn/the-6-pillars-of-the-aws-well-architected-framework/

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The optimal solution for a particular workload varies, and solutions often combine multiple approaches. AWS Well-Architected workloads use multiple solutions and enable different features to improve performance

5. Cost Optimization

The Cost Optimization pillar includes the ability to run systems to deliver business value at the lowest price point. You can find prescriptive guidance on implementation in the [Cost Optimization Pillar whitepaper](#).

Design Principles

There are five design principles for cost optimization in the cloud:

- Implement cloud financial management
- Adopt a consumption model
- Measure overall efficiency
- Stop spending money on undifferentiated heavy lifting
- Analyze and attribute expenditure

Best Practices

As with the other pillars, there are trade-offs to consider. For example, do you want to optimize for speed to market or

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6. Sustainability

The discipline of sustainability addresses the long-term environmental, economic, and societal impact of your business activities. You can find prescriptive guidance on implementation in the [Sustainability Pillar whitepaper](#).

Design Principles

There are six design principles for sustainability in the cloud:

- Understand your impact
- Establish sustainability goals
- Maximize utilization
- Anticipate and adopt new, more efficient hardware and software offerings
- Use managed services
- Reduce the downstream impact of your cloud workloads

Best Practices

Choose AWS Regions where you will implement workloads based on your business requirements and sustainability goals.

Use hybrid architectures to help you identify opportunities to meet sustainability goals. For example, code infrastructure

1. Which AWS value proposition allows customers to decrease the time to make resources available to administrators and developers from weeks to merely minutes?
 - Increase speed and agility
2. Which aspect of the AWS global infrastructure supports content distribution networks?
 - Edge locations
3. Which cloud computing concept refers to the ability to lower costs and raise efficiency when operating at a larger scale in comparison to operating at a smaller scale?
 - Economy of scale
4. Which cloud characteristic describes the ability to almost instantly provision and deprovision assorted cloud resources such as virtual instances, containers, appliances, database tables, and more?
 - Elasticity
5. Which well-architected pillar addresses the ability to support development and run workloads effectively, gain operational insight, and continuously improve supporting processes to deliver business value?
 - Operational Excellence
6. Which are design principles of the security well-architected pillar?
 - Keep people away from data
 - Implement a strong identity foundation
 - Protect data in transit and at rest
 - Enable traceability